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Deep Learning based Object Recognition for Robot picking task

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ABSTRACT

The primary objective of this research work is to converge into an accurate working algorithm for object recognition in a cluttered scene and subsequently helping the BAXTER robot to pick up the correct object in a cluttered environment. Feature matching algorithms usually fail to identify most of the object having no texture, hence deep learning methods have been employed for better performance. Although basic shallow Convolutional Neural Network (CNN) easily identifies the presence of an object within a frame, it is very difficult to localize the object location within the frame. This work primarily focuses on finding a solution for an accurate localization for robot grasping. In literature, YOLO (You Only Look Once) is found to be providing a very robust result on existing object recognition datasets. Due to high inaccuracy and presence of a huge redundant area within the bounding box, an algorithm was needed which will segment the object accurately and make the picking task easier. This was done through semantic segmentation using deep CNNs. Although time consuming, RESNET has been found to be very efficient as its post processed output helps to identify items in a significantly difficult task environment. This work has been done in light of recently held AMAZON robotics challenge 2017 where the robot successfully classified and distinguished a list of everyday items from a cluttered scenario. This work provides a performance analysis study comparing YOLO and RESNET based object recognition methods; and justifying through performance metrics (IOU - Intersection over Union) and ViG - Visual Grasping Score) that semantic segmentation (via RESNET) provides convenient results for robot vision module.

CCS CONCEPTS

• **Computing methodologies** → **Vision for robotics; Image segmentation; Object recognition; Supervised learning; Object recognition;**

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KEYWORDS

Convolutional Neural Network, Deep Learning, Object Recognition, Semantic Segmentation, YOLO, RESNET, Amazon Robotics Challenge

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1 INTRODUCTION

In the light of current advancement in deep learning, robot vision is not an exception. Many popular machine learning algorithms have already been proposed and implemented to solve intricate computer vision problems. The same has not been in the case of robot vision. Due to real time constraints and dynamic nature of environment such as illumination and processing power, very few algorithms are able to solve the object recognition problem at large. Robot vision is the ability of a robot to perceive its surroundings and interact according to the visual feedback in context of its environment. Robot vision extends the scope of computer vision in terms that it helps to fulfill the robot tasks in dynamic and real time conditions. Major goal of this project is to distinguish items in a cluttered scenario and produce a region with the recognizing item for better gripping or isolating it to its target position. It is to be noted that a significant amount image post processing has also been performed in order to produce the gripping region data in the item.

1.1 Objective

The overall objective involves picking of objects from a shelf where around 39 items are distributed more or less uniformly in different bins within the shelf. The primary task is to recognize a set of items from a clutter of objects assisted by the vision system and the robo-picker will put the items in the target bins. A secondary task involves the robo-picker to recognize a set of target items within a very large clutter of objects and put it back inside the movable shelf. Hence the overall objective for vision module is to recognize and localize a target item within the 2D pixel-space. The rest of the objectives are accomplished by the motion, control and gripper module of the robo-picker. It needs to be mentioned that the challenge involves distinguishing a set of previously known and unknown items. It is to be mentioned that the primary scope of this project is to detect the already known items. Detection of the unknown items have been solved with normal feature matching technique due to faster implementation, but with lesser efficiency [7]. In short the broad objective of the vision module is to distinguish a previously

known items within a dynamic cluttered scenario. The vision module receives a set of target objects and the task of the vision module is to detect whether the objects are present in the current frame of the robot camera. If the object(s) are present, the vision module will localize the detected items and preferably provide a metric of confidence. This work tries to explore different types of localization methods including bounding box output and semantic segmentation for better visual classification of target objects, eventually converging to a proper deep learning method via appropriate metrics.

1.2 Research Background

Ideal robot vision setup should be immune to dynamic changes from real time constraints such as illumination change, blurring and unexpected noise due to sudden and unexpected movement of the robotic system in action and its surrounding environment. In order to distinguish different objects, one needs to extract the features of the provided object followed by feature matching in the given environment. There have been many different feature extractors such as SURF [1] and SIFT [11]. Although qualitatively superior, they take a considerable amount of time for real time detection. There have been solution for them as well such as the ORB [17] extractor which provides sufficient keypoints for matching but is functionally inferior than SURF. For a large image database of different object, one needs to record all the keypoints and for each object and try to match it as per given problem.

Apart from this, another direct approach is to take the assistance of Machine Learning. In the past few years, several machine learning algorithms has been instrumental in recognizing and localizing objects in the pixel space. These include graphically drawing a rectangular grid indicating an object such as YOLO [14] or curving out the entire mask through semantic segmentation. Several such similar algorithms have been instrumental for the past few years. Special mentions include single shot detectors such as YOLO [15][21] and [9][20] as mentioned above. Apart from Deep residual learning [5],[18] Faster RCNN [16] has also been proposed for similar application for image recognition and classification [30]. Further bounding box detection scenarios has also been immensely implemented [8],[29],[28]. Image segmentation algorithms primarily include GoogleNet [24] and [5]. In this literature, we are more interested in finding a solution which provides a segmented mask corresponding to different objects. These are primary works on Convolutional Neural Networks (CNN) such as the end-to-end trained Fully Convolutional network [10] for image segmentation. Adding to this, DeepLab [2] is also a major contender for semantic image segmentation using deep convolution nets and fully connected CRFs. Adding to the list of end-to-end learning, DeconvNet [13] also receives a special mention. Although precise in segmentation, it has a time constraint even when trained with fast GPU based machines. Apart from this, several recent works involves the application of convnets to dense prediction problems, which also included semantic segmentation by [23]. The next section will focus on clarifying several popular terms in the field of deep learning based vision.

Application of semantic segmentation for robotics and autonomous system is not uncommon. We have seen some previous works on

segnet where it is applied on vision for autonomous cars [27]. Object recognition for robopicker has been employed before [6] via multi-class segmentation as well. This work uses a much more efficient algorithm of RESNET in order to identify and localize the objects. Apart from this, YOLO has also been applied for the first time to enhance the vision module for a robo-picker.

2 SYSTEM DESCRIPTION

2.1 Hardware

A more general setup included a robo-picker BAXTER (by *rethink robotics*) with specifically designed grippers for the picking and stowing task. Apart from this, the vision system is specifically maintained by input from Microsoft Kinect v2. The detection operation done by a nearby high power GPU enabled workstation. Apart from the PC screen, the segmented object can also be viewed on BAXTER screen.

Adding to this, a temporary artificial roof had also been added for preventing unwanted illumination for object recognition. Moreover, artificial lightning has also been introduced in order to provide a fixed illumination. Since only a 2D image recognition is required, input from only one KINECT is considered. It's worth mentioning that the workstation is Desktop which runs CUDA enabled NVIDIA GTX 1080 for high speed evaluation from the checkpoints provided from training.

2.2 Software

The robo-picker is operated by ROS (Py) module from the desktop workstation. Apart from this, OpenCV libraries and Python Image Libraries has also been extensively used for image post processing. Primarily, Tensorflow libraries has been extensively used for the implementation of YOLO and RESNET in Python. The training algorithm creates a checkpoint file after sufficient epoch run of the program and is transferred to the system workstation. The checkpoint file is nothing but the trained variables of the deep neural network.

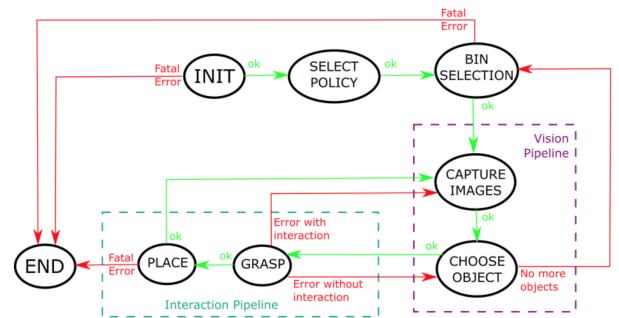


Figure 1: Total working pipeline for the robo-picker for object recognition, decision and picking.

3 CONVOLUTIONAL NEURAL NETWORK AND VISION

3.1 Deep Learning

Deep learning is nothing but a powerful set of techniques for efficient learning in neural networks. Deep neural networks are often much harder to train than shallow neural networks ([12]). In this section, we will briefly forage into the popular deep convolutional networks. These approaches have been tried before for the classical datasets and has given average to satisfactory results. After the recent onset of deep learning applications, these results seemed to achieve near perfection with deep neural network.

3.2 Convolutional Neural Network (CNN)

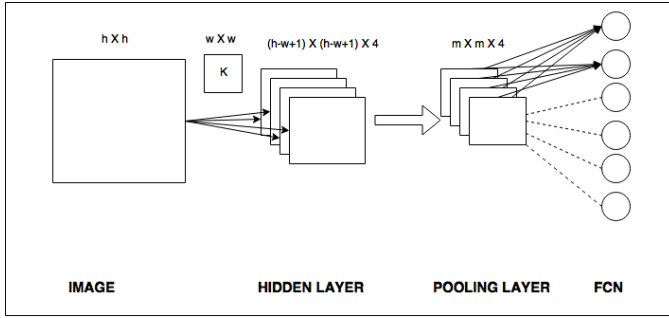


Figure 2: K denotes the convolution matrix kernel which slides over the Image input. Several of these kernels having different weight perform convolution and produce a set of hidden layer matrices, which are basically hidden neurons. Each of them detect a particular feature. Then it is followed by max pooling, and at the end it consists of a fully connected layer which finally provides the verdict for classification.

It is to be noted that the fully connected layers do not take into account the spatial nature of the image, which constitutes its pattern and structure. As rightly mentioned by Nielsen [12], it treats input pixels which are far apart and close together on exactly the same footing. Hence, a network is required which is suitable and well adapted for classification of images. This is exactly where CNNs come into action.

Each hidden neuron corresponds a bias and a matrix of weights associated with it. So for the entire single hidden neuron matrix, same weights and biases are used. For j,k th hidden neuron, the following equation provides the output [12]:

$$\sigma \left(B + \sum_{x=0}^C \sum_{y=0}^C w_{x,y} a_{j+x,k+y} \right) \quad (1)$$

σ denotes the activation function, B is the bias, w denotes the weight matrix and a denotes the input activation at a particular position of the image. For this particular property, the first hidden layer detects the same feature. It will detect that particular feature

irrespective of the position of the feature point present in the image. So even if a particular object is present in a different orientation, the network detects it anyways. Hence, it shows a translational invariance feature. These matrices which include weights and biases are also called kernels. The resultant map or matrix is called feature map. For better recognition through feature, several feature maps are required. The most prodigal advantage for this kind of network is that it requires much lesser number of parameters than its shallow neural network counterpart. Next, we have pooling layers. The pooling layer prepares a condensed feature map. It literally takes a small collection of pixels and summarizes its values. There are many techniques of pooling such as max-pooling and L2 pooling. At the end, there is fully connected layer (FC). It connects all the neurons from the last layer to each of the neuron in the FC. Basically, depending on the cumulative score, the particular pattern or object found is eventually classified.

3.3 Problem of Localization and initial approaches

A primitive CNN can only tell whether a particular object is present within the entire image or not. This is not what we are eventually looking for, but can be used as an initial step. So, the last layer basically provides us a score or probability of a particular object being present in the entire frame. Our final output should be exact co-ordinates of the location of a particular target object. It maybe in the form of a bounding rectangle localizing the object within the image or a segmentation mask showing precisely where the object is, on a pixel level accuracy. Before delving into that, a sliding window approach has been developed initially. Due to time constraints for multiple objects and due to very high rate of false positives, a simple sliding window approach has been found to be non-feasible in case of current scenario.¹

3.4 Single shot detector : YOLO

Single shot detectors usually provides a bounding box around the target object within the image frame. There are many such earlier works such as RCNN, Fast RCNN([16]), SSD [9] and YOLO [14]. In this section, we will be exploring the possibilities presented by YOLO (You only look once) detector. YOLO showed a much higher accurate prediction percentages than the other single shot detectors². This is the reason why we will explore our possibilities of using YOLO in the context of the Robo-picker.

Before going into details of YOLO, we will develop a general idea about how single shot detectors work. These methods usually localizes objects with regression. In short, instead of returning a class, it returns a bounding box (such as x,y,h,w). We train this system with an image having ground truth box as the label, and calculate the loss between ground truth and the predicted bounding box.

The main advantage of this detector is that it is much faster than its counterpart. The input image is generally converted into a 448 x 448 matrix. It is subsequently fed into the network and the output

¹A real time demonstration can be seen here: <https://www.youtube.com/watch?v=ncdJBpDF6RQ>

²Performance list of different bounding box detectors can be seen here <https://pjreddie.com/darknet/yolo/>

is filtered by NMS (Non-max suppression algorithm algorithm). YOLO consists of 9 convolution layers and after the 6th maxpool layer, the input image is gradually resized into 7×7 matrix. The final layer tensor dimension is determined as:

$$(S * S * (B * 5 + C)) \quad (2)$$

In short, the image after 9 layers of convolution and pooling, is resized into a dimension of $S \times S$ grid having B bounding boxes and C number of classes. So if we want to classify 40 objects and take $B = 5$ and $S = 7$ ([14]), then the dimension of the final tensor will be $7 \times 7 \times 50$.

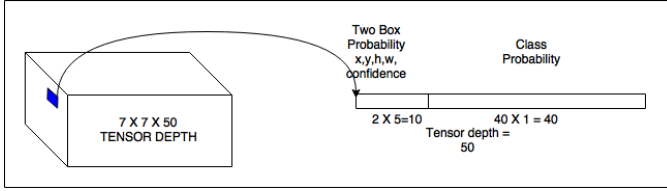


Figure 3: Final layer of YOLO. The image is squeezed into smaller matrix, but due to the depth of the tensor matrix, a significant amount of information is held intact.

The 7×7 tensor will now have two (variable C) box information (usually x,y,h,w) and confidence for each box. Apart from this, the rest of the 40 cells represents the class score per box. In short, each cell belong to a particular class and holds the probability information so that it can belong to a particular class. At the end, with the help of thresholding and NMS, we can filter out the invalid bounding boxes.

3.5 Semantic segmentation and RESNET

Segmentation involves detecting whether a particular pixel has any label associated with it or not. By evaluating that, one can easily determine a map corresponding to the area covering the particular object within the scene. It is to be noted that the last layer is converted into a convolutional layer with the help of up-sampling. Suppose an input image is eventually compressed into a single row vector CNN having a very deep tensor. At the last layer, denconvolution or transposed convolution is used to recover the activation positions which indicates the meaningful data residing within the given image. Other precise methods include 'extreme segmentation' ([19]) which is highly time consuming and it uses a method of unpooling for upsampling of image.

In this work, we will work on RESNET, a CNN based segmentation algorithm. It shows a very high efficiency on VOC 2007-2012 dataset [22] and this is the reason the RESNET has been chosen to be implemented on the current problem scenario [5].

The RESNET is basically a 150 layer deep network consisting of deep residual blocks. The actual idea is that it re-routes the input and adds the concept learned in the previous network. The idea is that the next layer would learn from both the inputs of the previous layer and the previous layer itself.

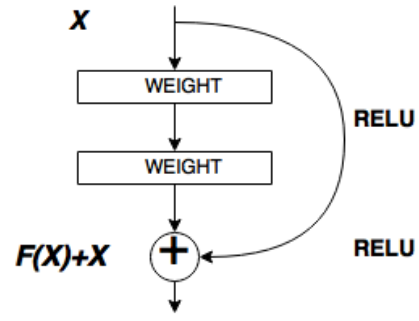


Figure 4: Basic building block of RESNET function

4 VISION SETUP FOR TRAINING

4.1 DATASET specification

The datasets consists of the main RGB image and its corresponding label. In case of YOLO, the label information came in the form of a four point coordinate within an image, and the object name. This information is stored in an XML file. The format for this dataset is exactly similar to PASCAL VOC dataset for localisation [4].

RESNET uses a labeled mask for object segmentation. The label is denoted by another RGB image having only single unique intensity shade around the segment space. The neural network takes in data in conjunction to the original RGB image.

The dataset was prepared manually for YOLO. The different images were localized with the help of software and the corresponding XML files were produced. In case of RESNET, the dataset was artificially generated³. Different objects were segmented out in a particular resolution and artificially placed in an expected background with known orientation, brightness and illumination. The corresponding label masks were also produced in sync with the RGB images. It is to be mentioned here that both the algorithms were tested on the same list of items in the cluttered background. But different labeling convention has been applied in the two different approaches. The application has been shown for different scenarios, maintaining an almost equivalent cluttering.

4.2 Dataset input and labels

The input data includes the target item in a cluttered scene, and a corresponding mask. The corresponding mask is a binary image having a blob around the particular item. The CNN is trained and it takes a cluttered scene as an input and produces a corresponding mask indicating the number of objects it has eventually found, in different colored shades. The dataset is produced by a repetitive algorithm which places different objects in different positions in the clutter. For a list of 39 items in the list, the total number of images used in this scenario is 10000. Correspondingly, 10000 binary mask images has also been produced for the purpose of giving the position to the CNN. The images of the objects were taken in separately, segmented and a transparent alpha channel was added as a background. It was then placed artificially in the cluttered scene, and a series of images were produced with multiple objects

³Our dataset generator: https://github.com/vasilya93/arc_image_generator

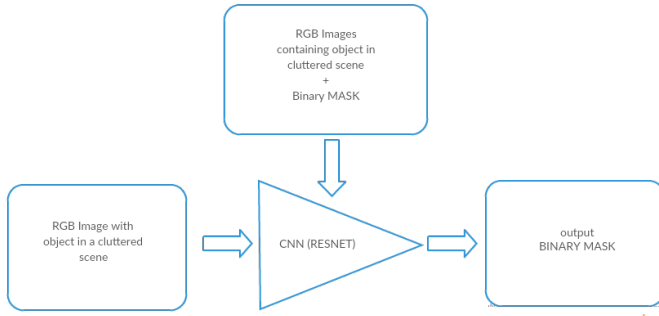


Figure 5: Graphical structure of the working dataset. It is to be noted that the Binary masks are labeled with respect to their intensity value and corresponding assigned item number.

in them, as shown in Figure 6. In short, items were placed artificially in different cluttered and non-cluttered backgrounds.

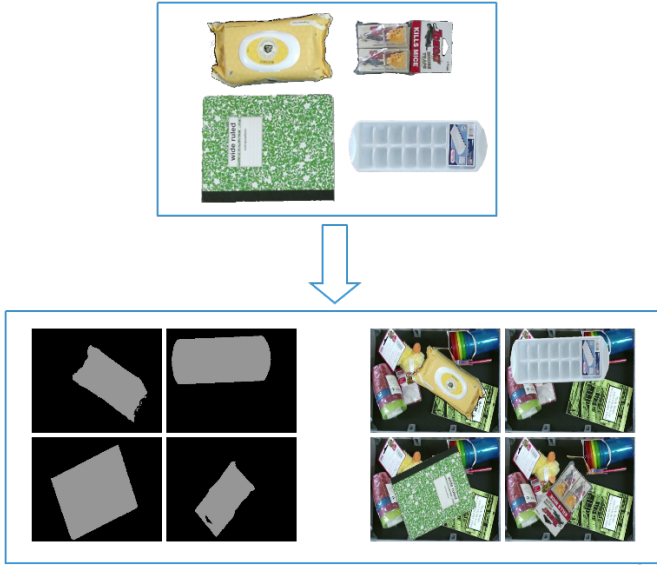


Figure 6: At first, only the object images are separated, by adding a transparent alpha channel. Then they are placed in artificially in cluttered scene as it's background. They also have a corresponding binary mask. The binary shades have different values depending on the item. This particular color value is the label of the image. Here it has not been shown for better comprehension. The last block is used as an input to the RESNET.

5 OBJECT RECOGNITION FOR PICKING

5.1 The challenge and the plan: revisited

The challenge is to detect objects from a very cluttered scenario and the sole objective of the vision module is to present a very accurate mask or a filter indicating the presence of the object within the



Figure 7: YOLO output of book. Although detection is done with a high accuracy, there is a lot of redundant space inside the ROI (Region of Interest). The purple region denotes the redundant space and any wrong item, if present might be picked up by the gripper due to detection of point-clouds not belonging to the item in the ROI.

frame. This mask will filter out the point-clouds which should be belonging to the objects only and this will be subsequently used by the grasping module to detect and grasp an object. The target is not to extract a region which will have the entire object in it, but to extract a region of interest which solely belongs to the object, without any redundant region surrounding it. Otherwise, in spite of detecting the correct item, the gripper might pick up the wrong object from the redundant region due to minute positional error of the gripper itself (see Figure 7).

5.2 YOLO: Observations

The tensor parameter was fixed according to the number of classes in the object list. The results are being depicted in Table 1. An alarming number of objects has been missed by the algorithm as shown by the results. It is to be noted that with a few exception, there is a huge redundant space within the bounding box which does not belong to the object, as cited in the previous section.

5.3 RESNET: Observations

A target object was chosen and the algorithm was run on an image frame containing the items. The resultant images show very promising results. The object space has been accurately extracted through semantic segmentation (see Table2). The grasping module now has a very accurate set of cloudpoints to chose from.

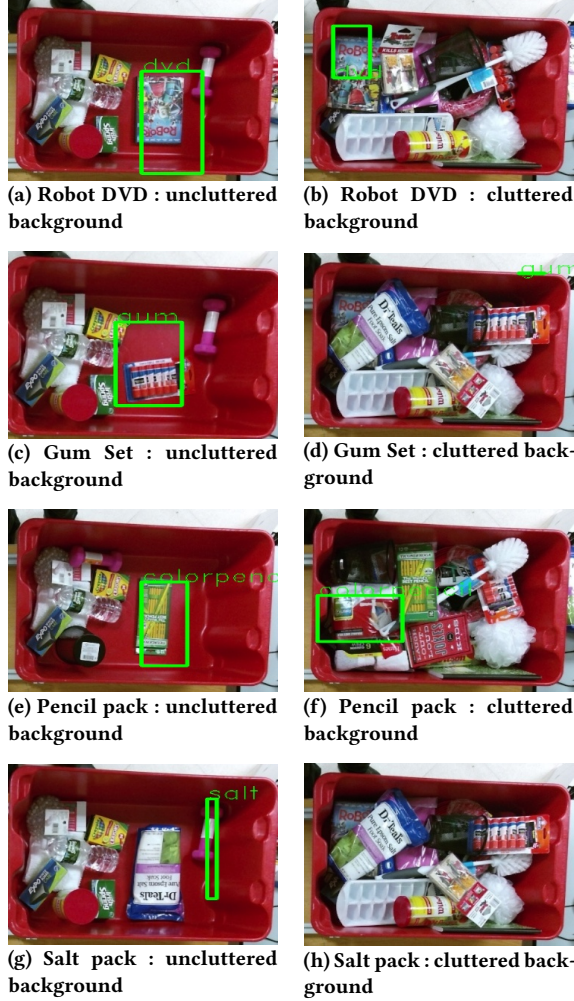
6 RESULTS AND EVALUATION

In the following sections we will discuss the results and the evaluation with respect to each algorithm. This will help us in concluding the practicability of using an algorithm for picking task by a robot.

6.1 Evaluation metric

In order to evaluate the two algorithms, we will be using Intersection Over Union (IoU) [25] and Visual Grasping score (ViG). The ViG factor is a novel proposal in this research work. Usually IoU has been used conventionally [3] to evaluate localization capability of an algorithm. But our target is to evaluate whether the information provided by the vision module is useful for grasp module or not. For this, we need to know what percent of the total detected pixels are originally within the actual object space? However small the space is, if the detected area lies within the ground truth; it should

Table 1: The YOLO Algorithm Produces a Bounding Box Surrounding the Box in an Uncluttered Background. But in Case of Cluttered Background, this Method Fails in a Significant Manner. It is to be Noted that this Result has been Produced after 57000 Iterations

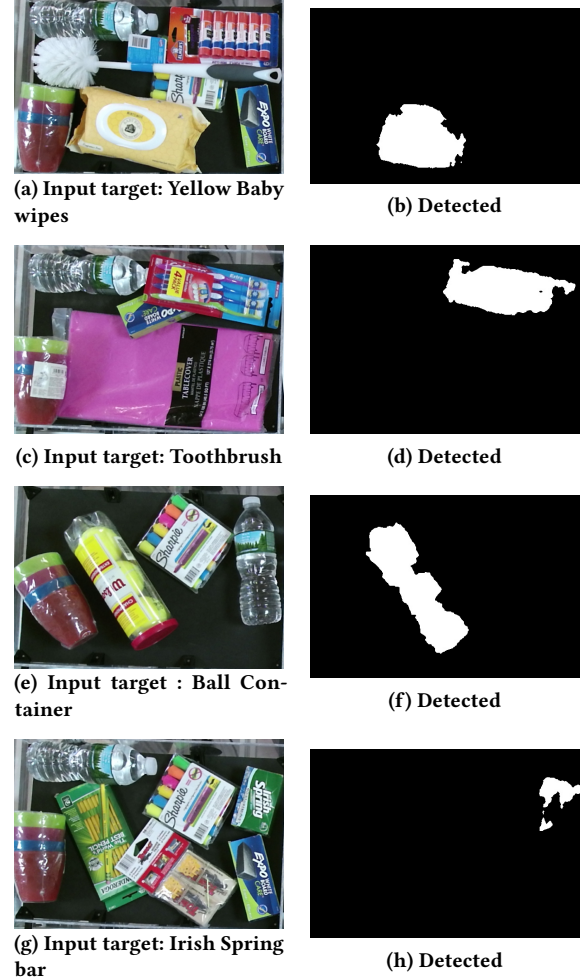


theoretically provide us full grasp-ability. Hence, keeping this particular condition in mind another metric has been developed which will provide a more accurate prediction of vision based grasping and subsequent picking. The ViG is nothing but the fraction of the detected area within the ground truth. Refer to Figure 8 for visual depiction of the metrics. Usually, it is expected that an algorithm must have an IOU greater than 0.5. It is to be noted that the factors ViG and IOU are not contradicting each other, but rather complementary. ViG enhances the detailed description of object recognition task with the help of IOU in this context.

6.2 Evaluation of YOLO

The YOLO algorithm has been tested with respect to the labeled ground truth. Refer Table 3 for sample examples on images. The

Table 2: Accurate Depiction of the Target Items Through Semantic Segmentation (RESNET). Total Iterations in This Case is 4500. This Seems to be a Concrete Option for Object Recognition for Picking Task



mean IOU after evaluating all of the items was found out to be 0.255. Additionally, The ViG score was found out to be 0.405.

6.3 Evaluation of RESNET

The RESNET algorithm has been tested in accordance to the labeled ground truth mask. Refer Table 4 for sample examples on images and the score. The mean IOU after evaluating all of the items was found out to be 0.49. Additionally, The ViG score was found out to be 0.94.

6.4 Discussions

It is to be noticed that the primary focus of this work is not to make a comparative analysis of the given algorithms, but to highlight the way how we reached a solution for our own requirement and functionality. The target of this work was to produce a very accurate

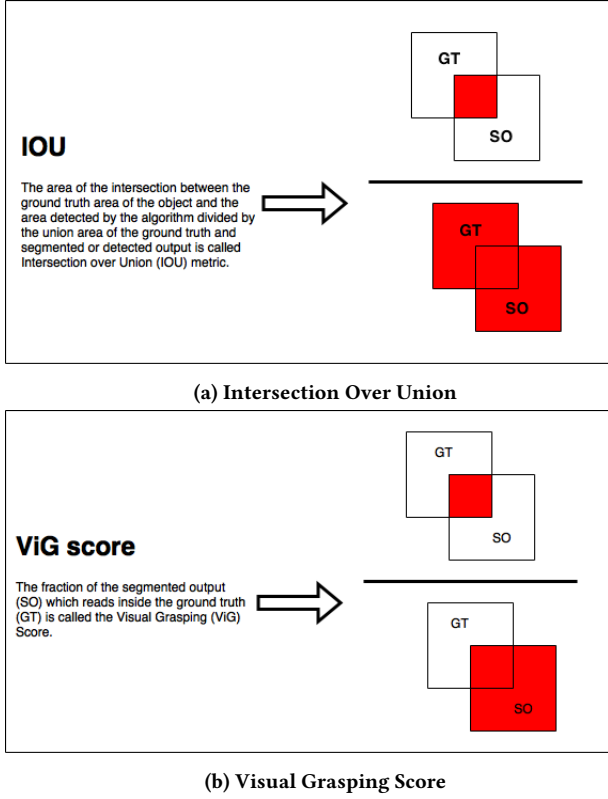


Figure 8: Aforesaid figures show metrics used for the evaluation of the algorithms. It is to be noted that our primary concern is the ViG score. Higher score means that more number of pixels are inside the particular object in the image, meaning easy grasp probability by the gripper module.

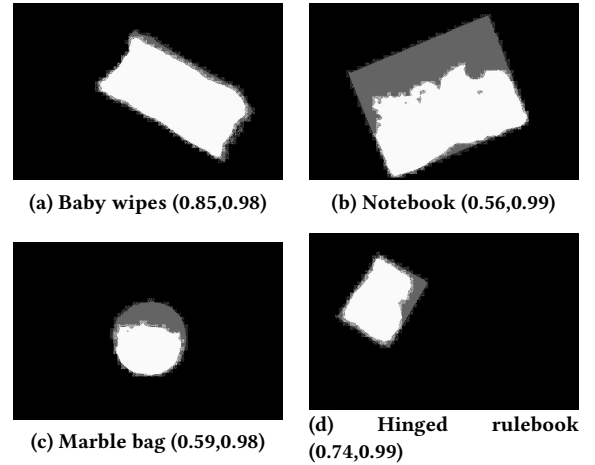
map of the point-clouds which completely belongs to the particular object. In our case, we have utilized RESNET in order to do that as it doesn't produce a redundant area within a bounding box like YOLO.

We can see a significant jump in performance when we apply a semantic segmentation network for vision for picking task (see Table 5). In this case, it is to be noted that we are more interested in the pixels that belong to the original object and it is important that the pixels belong to the original object only. This parameter is shown precisely by ViG metric. The IOU, although a very significant metric, doesn't show the full picture. It is not being proposed to choose a metric over the other, but to use the metrics in congruence with the other as they convey different application information. The IOU displays information about the vision performance, whereas ViG displays theoretical ability of the grasping module to use the visual information for efficient grasping. Both the metrics are interrelated, where ViG can be seen as specialization of the IOU metric itself.

Table 3: Results Produced by YOLO is Mostly Unreliable for Object Detection. In Some Cases, as Shown Above, it Detects the Wrong Object and Sometimes Outside the Coordinates of the Frames. The Results Shown here were Produced after 57000 iterations. It is to be Noted that the Green Bounding Box Denotes the Ground Truth Used as a Label, and the Red Box Shows the Output from YOLO



Table 4: IOU and ViG Score of RESNET Segmented Mask for Different Objects. The Gray Part Denotes the Part of Ground Truth Which is not Detected. The White Part Shows the Detected Area by RESNET



7 CONCLUSIONS AND FUTURE WORK

7.1 General Conclusions

It can be safely concluded from the experimentation that using a semantic segmentation based algorithm is a better proposition than to use a localization method which produces bounding box as an output. As shown before, apart from the detection error we have a redundancy error. In the later case, unwanted pixels are also selected which do not belong to the object at all. This problem has

Table 5: Mean IOU And ViG Score Of RESNET And YOLO Algorithm

Algorithm	IOU	ViG
YOLO	0.255	0.405
RESNET	0.49	0.94

been solved by semantic segmentation. In RESNET based semantic segmentation, the objects have been segmented out very carefully with pixel level accuracy and is free of rectangle shape related error which previously produced the redundancy. The detection accuracy almost doubled when compared to the single shot detector algorithm (YOLO). Hence it can be concluded that the current research supports that semantic segmentation (implemented as RESNET) performs better than bounding box (implemented as YOLO). Hence in an application like this, where it is needed to select precise 3D point-clouds of objects mostly having irregular shapes, semantic segmentation based methods provide a better filter for segmenting out point-clouds belonging to the objects at large, especially in a cluttered background. The aforesaid work also shows segmentation output of most of the items graphically. In most of the cases the segmentation mask do not cover the entire object, but the segmentation mask almost solely belongs to the object itself which is shown by the ViG score. This gives the correct cloud-points to the grasp module.

7.2 Future Work

There is a huge scope for future work for improving the application base. It is to be mentioned that the primary focus for the future work is not only to improve the algorithm itself, but to improve upon the usability and application itself. We can simply provide a very short list of probable and promising future aspect of continuing this work:

- Fast data acquisition and training
- Compatible with portable devices
- Grasp enabled multi-labeling
- Implementation in other common application

With the advent of faster GPUs, RESNET can be implemented in a very fast manner. In case of unknown items, an advanced dataset acquisition module needs to be developed which will automatically extract the alpha channel based information, subsequently producing the artificial dataset and the masks, train it on the algorithm via transfer learning and implemented on-the-go. Apart from making the algorithm faster, focus should be made on implementing it on unknown items and implementation in a quick manner. This constitutes not only the algorithm itself, but the peripheral dataset acquisition module as well.

There are numerous application for semantic segmentation based analysis. Segmentation of scene for self driving cars, human pose estimation ([26]) are one of the very few sections where multi-labeling and semantic segmentation based task is utilized for practical application based scenarios.

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